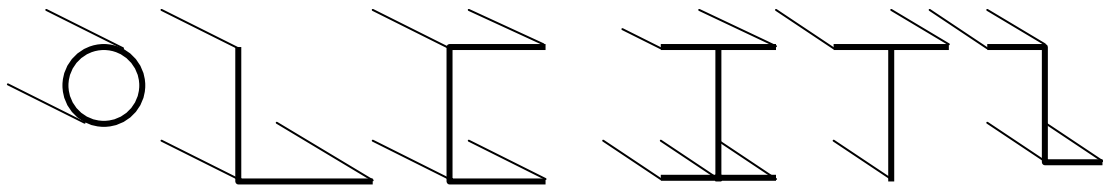


Properties of Sections

In order to start to consider simple problems in bending we need to know a bit about properties of simple sections. Due to the wide range of sections in use it is necessary to classify them into two main types, namely simple symmetrical sections and more complex unsymmetrical and irregular shaped sections. Simple sections will be dealt with here and complex sections will be introduced later in the course.

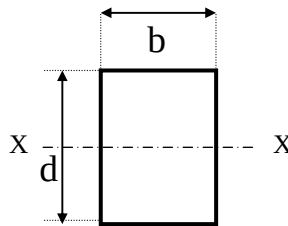


Tube Angle Channel I-Section T-Section Z-Section

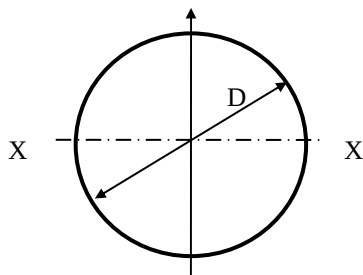
Second Moment of Area, I for standard cross-sections

Rectangle

$$I_{xx} = \frac{bd^3}{12}$$

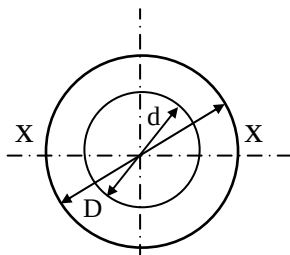


Circular section



$$I_{xx} = \frac{\pi}{64} D^4$$

Hollow Circular Section

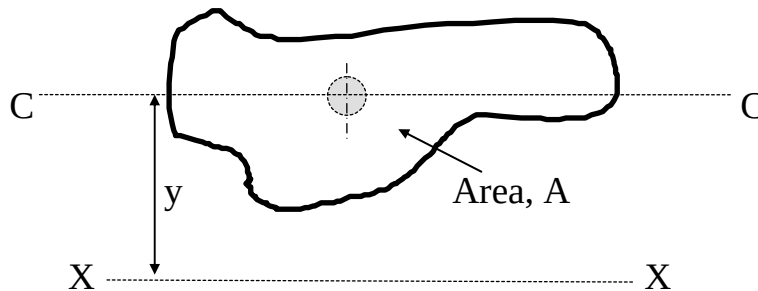


$$I_{xx} = \frac{\pi}{64} (D^4 - d^4)$$

Where D is outside diameter; d is inside diameter

Parallel Axis Theorem

The second moment of area, of an area about an axis, is equal to the second moment of area of the area about an axis passing through its centroid plus the product of the area and the square of the distance between the two axes.



For the area, A, shown above:

$$I_{xx} = I_{cc} + Ay^2$$

Second moment of area is analogous to the second moment of mass (or inertia) in Dynamics.

Term ‘Moment’ usually associated with forces and distances - here considering areas and distances.

How to calculate Second Moments of Area

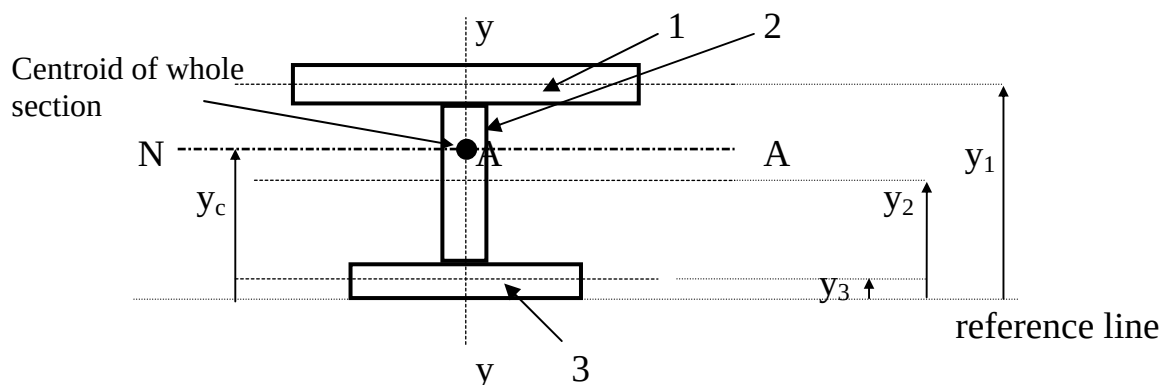
(for simple sections with an axis of symmetry)

- Typical sections you will meet have yy as a line of symmetry:



- First we need to determine the position of the centroid, then calculate I .
- Split it into rectangles 1, 2 and 3.
- Draw on a reference line at the bottom of the section.

Assume the centroid lies a distance y_c up the yy axis, measured from the reference line.



We find y_c by constructing the following table:

section	b	d	$A=b \times d$	y	$A \times y$
1	b_1	d_1	$b_1 \times d_1$	y_1	$b_1 \times d_1 \times y_1$
2	b_2	d_2	$b_2 \times d_2$	y_2	$b_2 \times d_2 \times y_2$
3	b_3	d_3	$b_3 \times d_3$	y_3	$b_3 \times d_3 \times y_3$
Total Σ			ΣA		ΣAy

Then using:

$$y_c = \frac{\sum Ay}{\sum A}$$

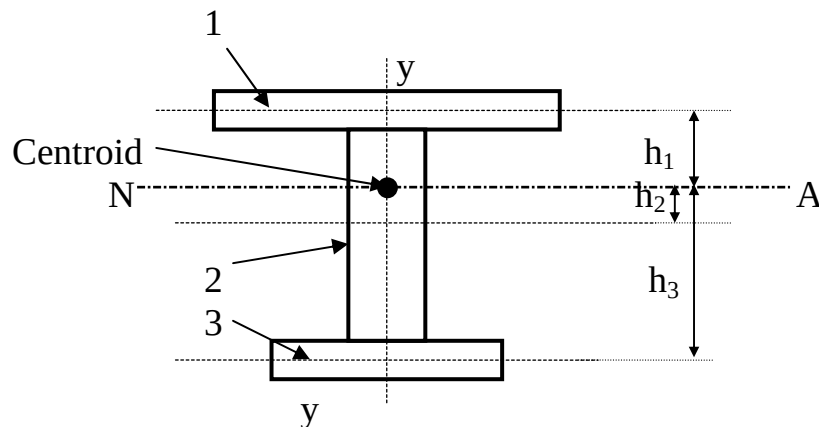
Determine y_c .

Now calculate the Second Moment of Area, I:

Add to the right hand side of your table more columns (or draw separately, if no room).

h_i	$A \times h_i^2$	I_{cc}
h_1	$A_1 \times h_1^2$	$(b_1 \times d_1^3)/12$
h_2	$A_2 \times h_2^2$	$(b_2 \times d_2^3)/12$
h_3	$A_3 \times h_3^2$	$(b_3 \times d_3^3)/12$
	$\sum Ah^2$	$\sum I_{cc}$

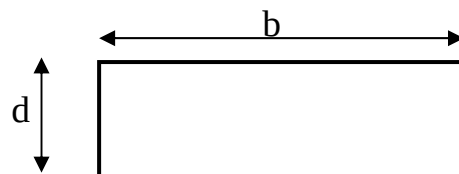
In this table ' h_i ' is the distance from the centroid of the whole section, to the centroid of the rectangle we are considering. This is shown below: -



Also in this second table the Ah^2 column is simply the area of the rectangle multiplied by the respective values of h^2 .

The I_{cc} column is the Second Moment of Area for each individual rectangle about its own centroid

e.g. For the rectangle shown:



$$I_{cc} = bd^3/12$$

Finally, if you sum both the Ah^2 and I_{cc} columns, you get I_{NA} from:-

$$I_{NA} = \sum I_{cc} + \sum Ah^2$$

Summary of steps

To determine y_c and I_{NA} follow these steps:

1. Split section into rectangles, then number them.
2. Draw section with reference line at bottom.
3. Draw table to calculate $\sum A$ and $\sum Ay$
4. Use $y_c = \frac{\sum Ay}{\sum A}$ to calculate y_c .
5. Draw 3 more columns on table, h , Ah^2 and I_{cc} . Remember: h is measured from centroid to centroid.
6. From table find $\sum Ah^2$ and $\sum I_{cc}$.
7. Use $I_{NA} = \sum I_{cc} + \sum Ah^2$

Combined Bending and End Load

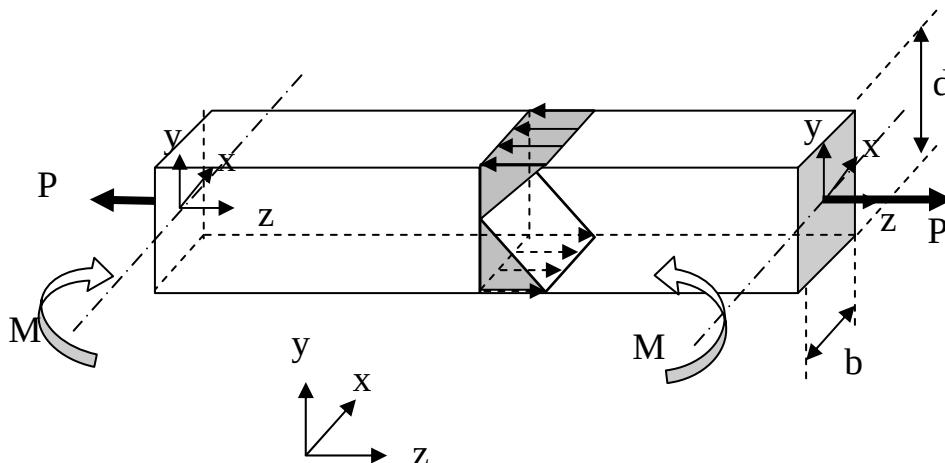
A number of situations arise in practice where a member is subjected to a combination of bending and longitudinal load (a thrust or a pull).

We are concerned with combined bending and end loading, and in the cases where it is an end thrust we will assume that this load does not lead to overall instability of the member. Instability and buckling will be dealt with later on.

Problems of this type can mostly be dealt with by 'superposition' of the individual components of stress to give resultant values.

If a member carries a thrust or pull, direct longitudinal stresses are set up; if a bending moment is now superimposed on the member at some section, further longitudinal stresses are induced.

For example, consider the uniform rectangular section beam which is subjected to a tensile end load, P and a bending moment M about the xx axis:



If the end load P acted alone:

$$\sigma_z = \frac{P}{A} \quad ; \quad \text{Where} \quad A = bd$$

If the moment M acted alone: $\sigma_z = \pm \frac{My}{I}$

By superposition the resultant stress due to P & M :

$$\sigma_z = \frac{P}{A} \pm \frac{My}{I} = \frac{P}{bd} \pm \frac{12 My}{bd^3}$$

Single Axis Bending

The maximum stress in a section subject to direct axial loading together with an externally applied moment acting about either the xx or the yy axis is given by:

$$\sigma = \frac{P}{A} \pm \frac{M_x y}{I_{xx}} \quad M_x \text{ acting about the xx axis}$$

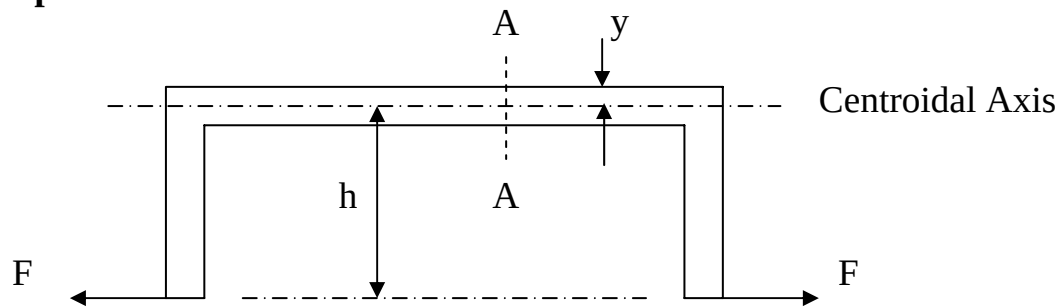
or

$$\sigma = \frac{P}{A} \pm \frac{M_y x}{I_{yy}} \quad M_y \text{ acting about the yy axis}$$

where:

A	=	Area of cross-section under load
P	=	Direct axial load
M_x	=	External moment about the xx axis
M_y	=	External moment about the yy axis
I_{xx}	=	Second moment of area about xx axis
I_{yy}	=	Second moment of area about yy axis
x	=	Distance from yy axis
y	=	Distance from xx axis

Example 1



Determine the stress resultant stress at section AA.

At section AA we have 2 effects:

1. Direct stress (F/A)
2. Bending stress (My/I)

Where: y and I depend on the section at AA

$$\sigma_{\text{TOTAL}} = \sigma_{\text{DIRECT}} + \sigma_{\text{BENDING}}$$

$$= (F/A) + \frac{(F \times h) \times y}{I}$$

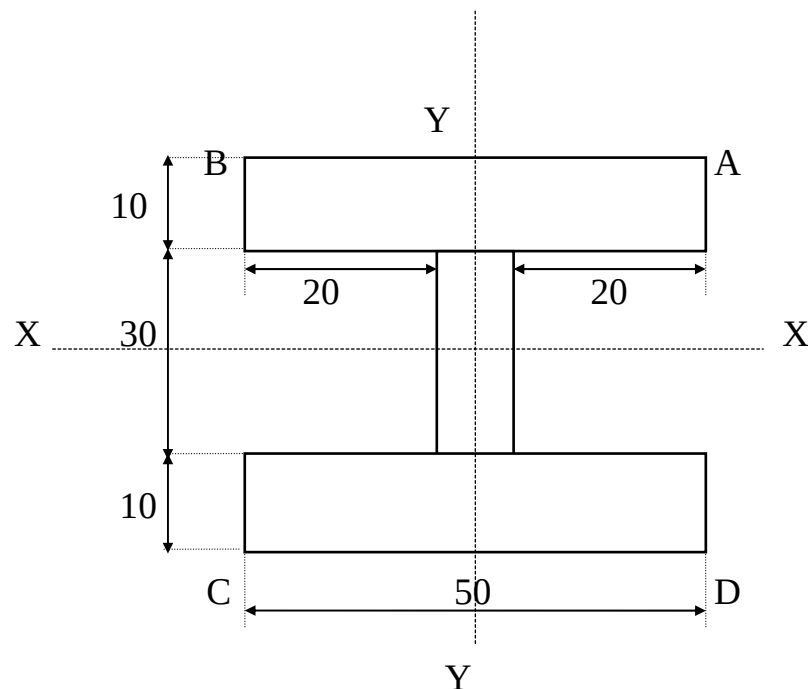
Bi-axial Bending

The maximum stresses in a section subject to direct axial loading together with moments M_x AND M_y acting together:

$$\sigma = \frac{P}{A} \pm \frac{M_x y}{I_{xx}} \pm \frac{M_y x}{I_{yy}}$$

Example 2

All dimensions in mm



At a particular cross-section a beam is subjected to :

- A tension of 50 kN acting through the centroid.
- A bending moment about the xx axis of value 0.8 kN.m which causes tension along face AB.
- A bending moment about the yy axis of value 0.5 kN.m which causes tension along face BC.

Calculate the stresses at the 4 corners A, B, C and D.

Note:

- The section is symmetrical.
- The two axes of symmetry passing through the centroid are co-linear with the specified xx and yy axes.
- Assume the material behaves in a linear-elastic manner.

Firstly we need to calculate A, I_{xx} and I_{yy}

Then we can begin to determine the stresses on the cross section.

$$a) \quad (P/A) = \frac{50 \times 10^3}{1300} = +38.46 \text{ N/mm}^2 \text{ Tensile}$$

$$b) \quad \frac{M_x \cdot y}{I_{xx}} = \frac{0.8 \times 10^6 \times 25}{0.431 \times 10^6} = \pm 46.40 \text{ N/mm}^2 \text{ Tensile on AB}$$

$$c) \quad \frac{M_y \cdot x}{I_{yy}} = \frac{0.5 \times 10^6 \times 25}{0.211 \times 10^6} = \pm 59.24 \text{ N/mm}^2 \text{ Tensile on BC}$$

Therefore from:

$$\sigma = \frac{P}{A} \pm \frac{M_x y}{I_{xx}} \pm \frac{M_y x}{I_{yy}}$$

$$\sigma_A = 38.46 + 46.40 - 59.24 = 25.62 \text{ N/mm}^2$$

$$\sigma_B = 38.46 + 46.40 + 59.24 = 144.10 \text{ N/mm}^2$$

$$\sigma_C = 38.46 - 46.40 + 59.24 = 51.30 \text{ N/mm}^2$$

$$\sigma_D = 38.46 - 46.40 - 59.24 = -67.18 \text{ N/mm}^2 \text{ (comp)}$$

Eccentrically Loaded Sections

Combined end load and single axis, and even bi-axial, loading conditions are often easy to identify as the moment arm of the applied force may be large. A less obvious example is where the force appears to be along the axis but is offset from the centroid.

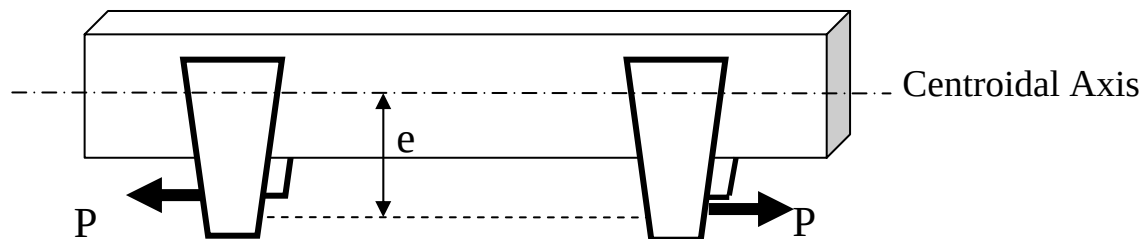
If a load P acts at an eccentricity, e from the centroidal axis of a section, then that loading is equivalent to:

- A direct load P , acting axially through the centroid of the section;

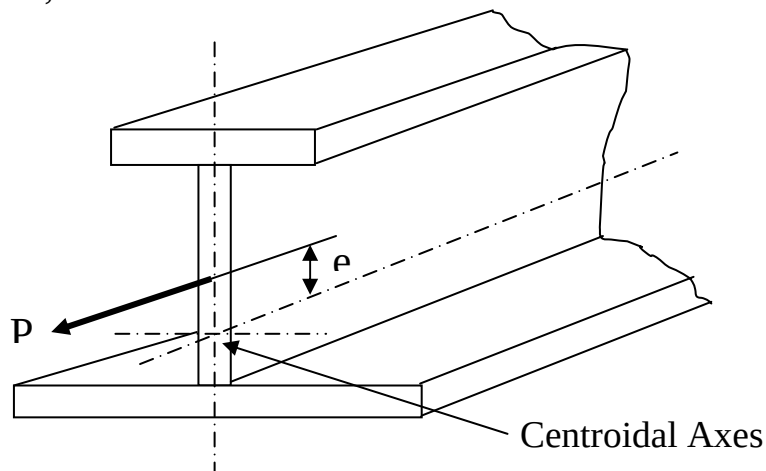
and

- A moment of the value $M = (P \times e)$ acting about the axis of bending which passes through the centroid of the section.

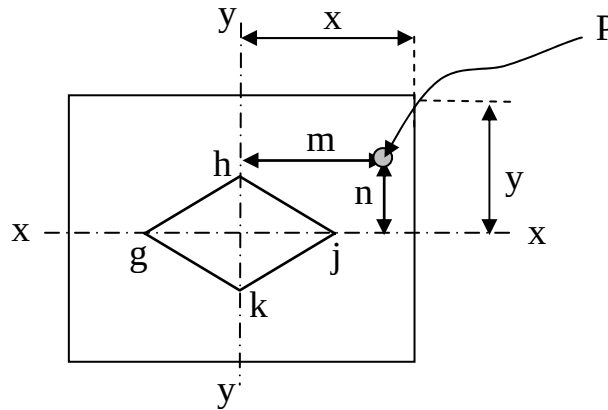
An obvious example is when loads are offset by means of brackets:



A less obvious example is where the force appears to be along the axis but is offset from the centroid, an amount 'e'.



Core of a Section



The load shown above is equivalent to the load on the centroid, together with a moment $P.m$ about yy and $P.n$ about xx .

$$\therefore \text{Stress at any point} \quad \sigma = \frac{P}{A} \pm \frac{P.m}{I_{yy}} \cdot x \pm \frac{P.n}{I_{xx}} \cdot y$$

The condition for no tension is that the compressive stress due to P is greater than or equal to the maximum tensile bending stresses set up by moments $P.m$ and $P.n$.

IF there is no change in the nature of the stress in the section (e.g. still tensile even with the offset loading), then the load must be applied within the core of the section **g h j k**, where:

For a rectangular section:

$$\mathbf{g j} = 1/3 \text{ of horizontal axis}$$

$$\mathbf{h k} = 1/3 \text{ of vertical axis}$$

Example 3

- Find the core of the rectangular section shown:

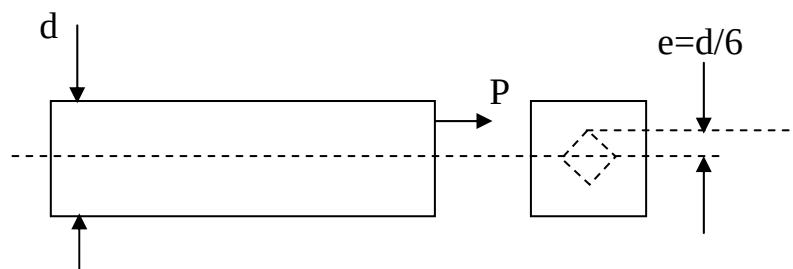
$$\sigma = \frac{P}{A} \pm \frac{P \cdot e}{Z} \quad \text{where } Z = \text{Section Modulus} = I/y_{\max}$$

$$A = bd \quad \& \quad Z = \frac{bd^3 \times 2}{12 \times d} = \frac{bd^2}{6}$$

$\therefore$ if $\sigma = 0$ on the edge,

$$\therefore 0 = \frac{P}{bd} \pm \frac{Pe}{bd^2 / 6}$$

$$\therefore e \leq \frac{d}{6}$$

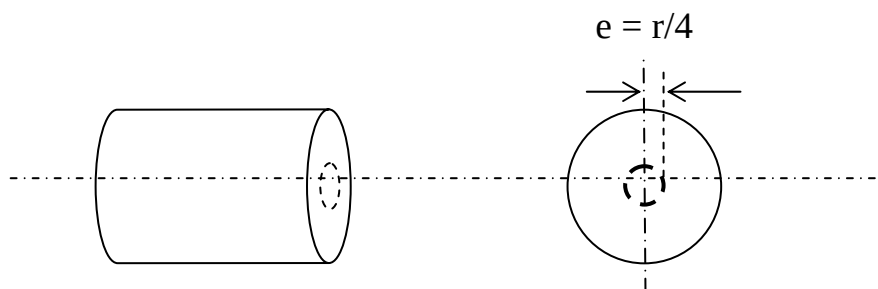


- Find the core of the circular section shown:

$$\sigma = \frac{P}{\frac{\pi D^2}{4}} \pm \frac{Pe \frac{D}{2}}{\frac{\pi D^4}{64}}$$

If $\sigma = 0$ on the edge,

$$\therefore e \leq \frac{D}{8} \text{ or } \frac{r}{4}$$

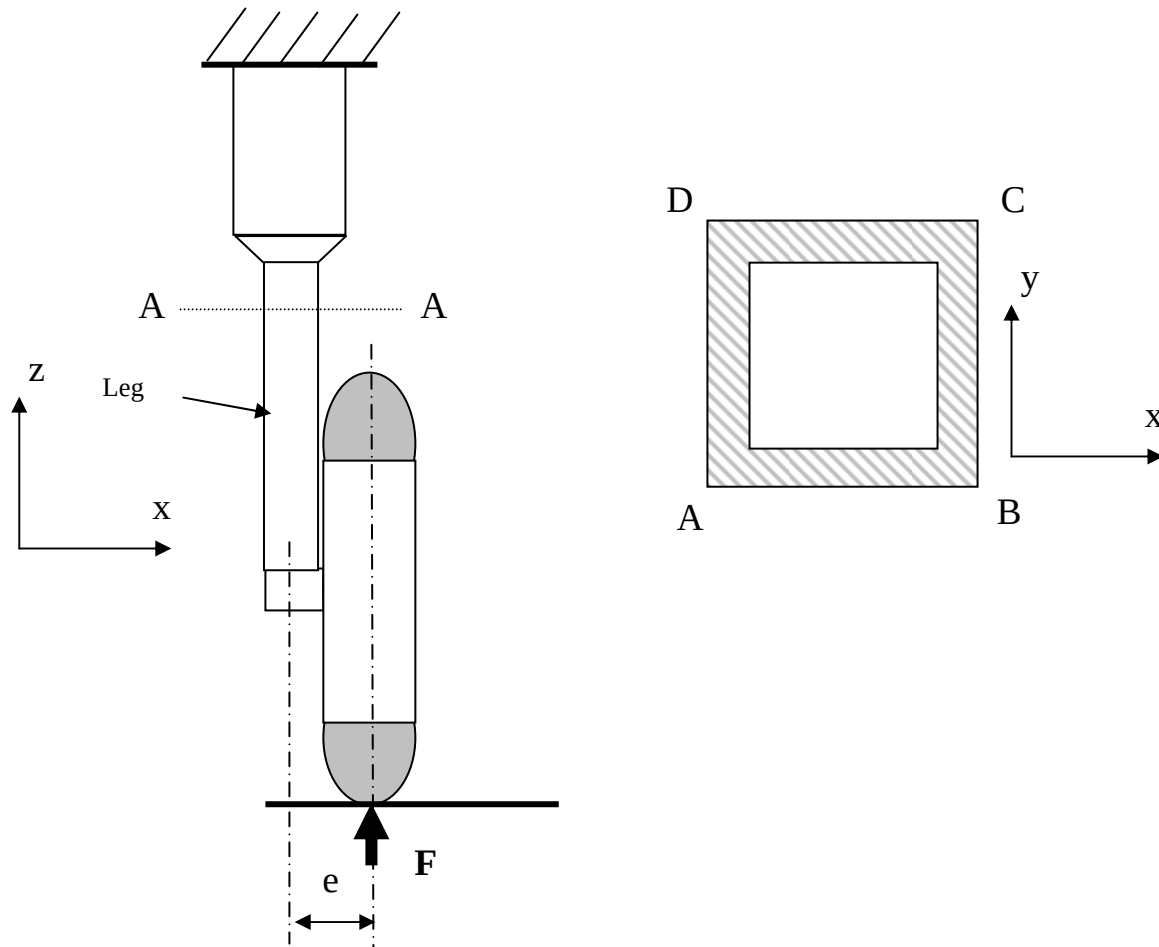


If load is applied within, the core stresses will have the same sense throughout the section.

Tutorials

Question 1

A simplified front wheel arrangement for an aircraft is shown below. The leg is made from hollow square tube with outside dimensions 50mmx50mm and with a thickness of 6mm.



Find the magnitude and position of the maximum tensile and compressive stresses at section AA, for a wheel offset, e of 0.2m and a force F of 10kN.

Sketch the stress distribution at section AA.

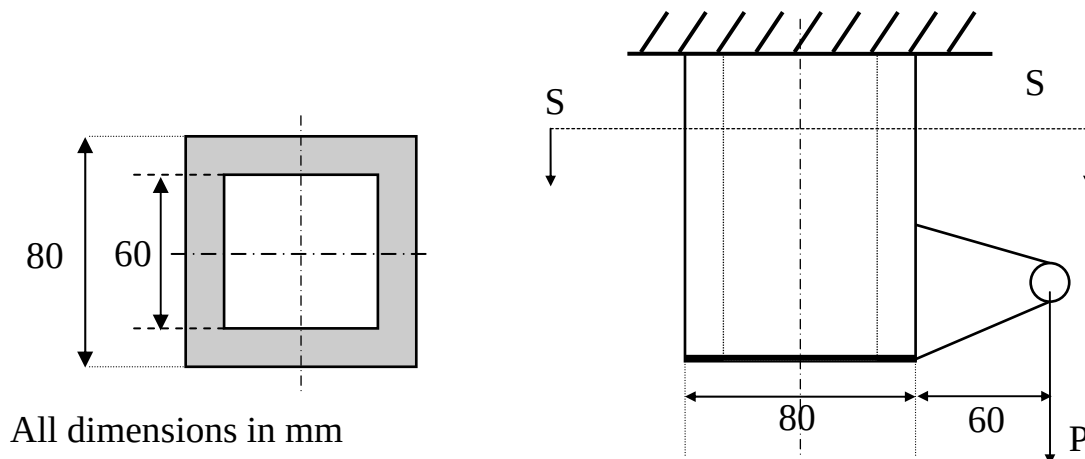
If the maximum stress at the section is not to exceed 150MN/m^2 , determine the maximum allowable wheel offset, e .

Consider why the resultant stress at any point can simply be determined by application of a simple superposition technique.

Question 2

A short box beam carries an eccentric load P of 72kN. Assume a uniform wall thickness of 10mm and box cross section of 80 x 80 mm.

Determine the values of the maximum tensile and compressive stresses on a cross section, such as SS. Show the stress distribution on a diagram.



All dimensions in mm

Question 3

A component has to be curved in one plane in order to allow the required clearance around adjacent components. Estimate the maximum tensile and compressive stress in the link under the action of an end thrust of 2500N.

